



How To Test Thermal Imaging Cameras

Thermal sensitivity is the ability to distinguish tiny differences in temperature. It has a high impact on detection performance

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A thermal camera measures temperature of a surface by interpreting density of an infrared (IR) signal emanating from that surface and reaching the camera itself. Every object or environment includes materials with different levels of emissivity and reflectivity, and understanding these properties is vital for accurate temperature measurements. The ability to measure the surface temperature from a distance not only helps ensure safety of the person operating the camera but also ensures that the item being measured is not influenced by the measuring instrument.

Thermal imaging works on the principle of measuring reflected IR radiation from an object or environment to detect flaws or different objects. IR waves are a part of electromagnetic radiation whose wavelength is between 700nm and 1mm. However, there are three spectral ranges that are considered important for thermal sensor applications: 1 μ m to 3 μ m short-wave IR, 3 μ m to 5 μ m middle-wave IR, and 8 μ m to 14 μ m long-wave IR.

An IR camera used in thermal imaging needs to be tested for sensitivity to the tem-

perature difference between an object and its background. This requires simulation of the object and its background by placing a target in front of the emissive surface of a blackbody to have a simultaneous and accurate measurement of both.

An IR target is a thin sheet with high emissivity coating and stuck patterns. The object is made by the blackbody seen through holes, while the background is the solid part of the sheet. Consequently, temperature of the object is temperature of the blackbody, while temperature of the background is measured by inserting a temperature sensor into the mount of the target.

Different targets for different tests

The most frequent targets for usual tests of an IR camera are: pinhole target for optical axis alignment of optronic system, square or rectangle target for field-of-view (FOV) measurement, grid holes for distortion image measurement, half-moon target for modulation transfer function (MTF) calculation and four-bar target for thermal resolution measurement. Noise tests usually do not require any target.

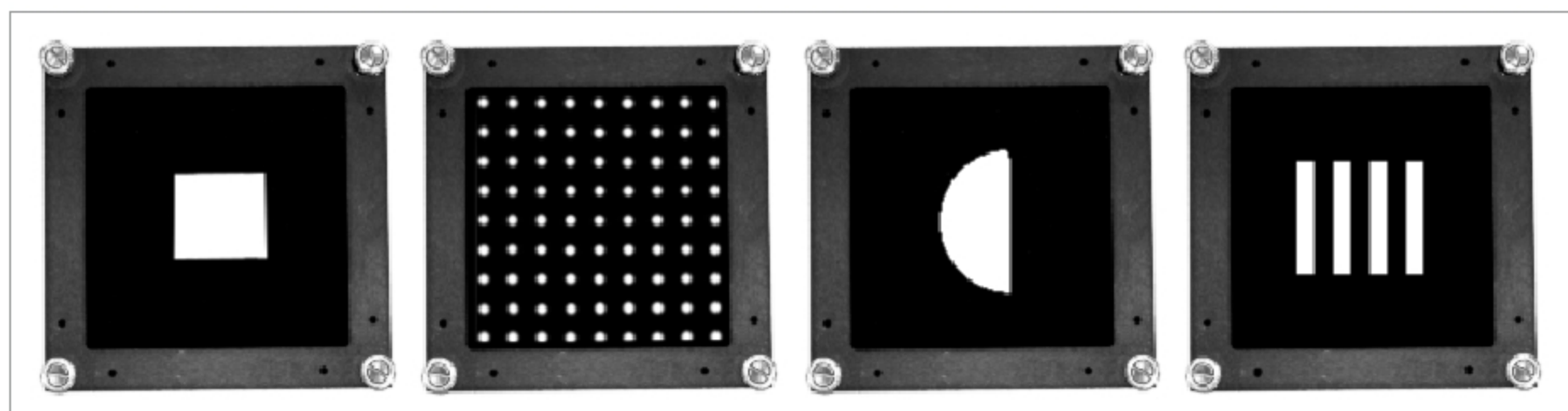


Fig. 1: Different targets for different tests (Credit: www.hgh-infrared.com)